C1.1 Enzymes and metabolism

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c1-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 396–404 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

METABOLISM

C1.1.11 Intracellular and extracellular reactions

p. 396

Most enzymes work **inside** the cell that made them, for example in glycolysis (cytoplasm) and the Krebs cycle (mitochondrial matrix). Some are secreted by exocytosis and work **outside** the cell, such as the digestive enzymes that break down food in the gut.

C1.1.12 Metabolism releases heat

pp. 396–397

No energy transfer in metabolism is 100% efficient, so heat is always released. Mammals and birds use this heat to keep a constant body temperature, and can raise it by shivering or by brown fat, which releases energy from respiration as heat.

C1.1.13 Linear and cyclical pathways

pp. 397–398

Metabolic pathways are series of enzyme-catalysed steps. In a **linear** pathway, such as glycolysis, a substrate is converted to an end product. In a **cyclical** pathway, such as the Krebs cycle and the Calvin cycle, the last step regenerates the first compound, so a small amount of it can process many molecules.

INHIBITION

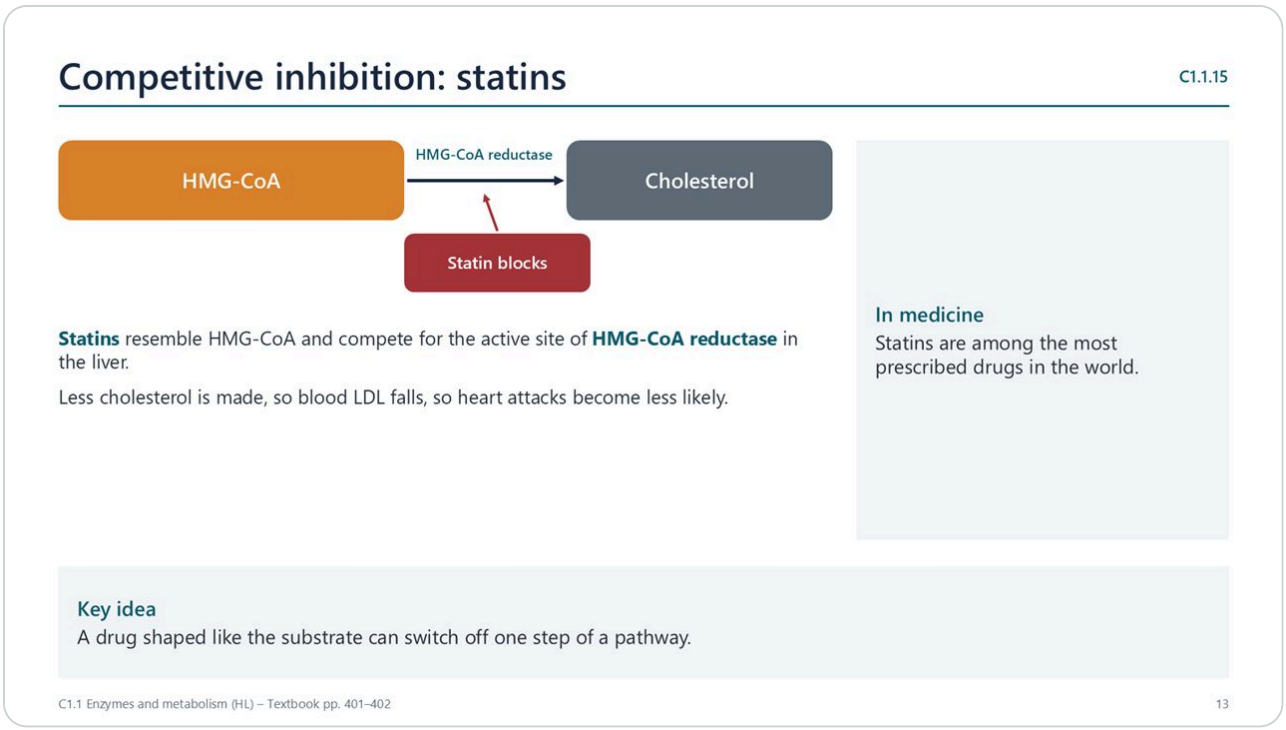
C1.1.14 Allosteric (non-competitive) inhibition

pp. 398–399

An **allosteric site** is a binding site away from the active site. Only specific molecules fit it. Binding changes the shape of the enzyme so the active site can no longer catalyse the reaction. Binding is reversible, and adding more substrate does not overcome the inhibition, so the maximum rate is lowered.

C1.1.15 Competitive inhibition

pp. 401–402



A competitive inhibitor resembles the substrate and binds reversibly to the **active site**. At high substrate concentration the substrate outcompetes it, so the maximum rate can still be reached. **Statins** inhibit HMG-CoA reductase in this way, reducing cholesterol synthesis in the liver and lowering LDL in the blood.

C1.1.16 Feedback inhibition

pp. 402–403

In bacteria and plants, **isoleucine** is made from threonine in a pathway of several steps. When isoleucine builds up, it binds to the allosteric site of the first enzyme, **threonine deaminase**, and slows the whole pathway. As isoleucine is used, it leaves the enzyme and production restarts. This negative feedback avoids wasting resources.

C1.1.17 Mechanism-based inhibition: penicillin

pp. 403–404

Bacteria use **transpeptidases** to cross-link peptidoglycan in their walls. Penicillin enters the active site, its four-membered ring opens and it forms a **covalent bond** with the enzyme, permanently inactivating it. Bacteria with a changed transpeptidase, whose active site binds penicillin poorly but still cross-links the wall, are **resistant**.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission